Do Beautiful Parents Have More Daughters?[1]

A Bayesian Case Study in Small-Effect Inference

After Gelman & Weakliem (2009), Kanazawa (2007), and Gelman, Hill & Vehtari (2020) §9.4

Lecture notes for class use.

Prerequisites. Probability and conditional probability; Binomial and Normal distributions; maximum-likelihood estimation; basic Bayesian inference (likelihood, prior, posterior); the Beta distribution as a conjugate prior.

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1 Background and data

1.1 The scientific claim

Evolutionary psychologist Satoshi Kanazawa proposed the *generalised Trivers–Willard hypothesis* (gTWH): parents who possess a heritable trait that increases the reproductive success of daughters more than sons should, over evolutionary time, produce more daughters. His specific prediction was that physically attractive parents, whose looks benefit daughters more than sons reproductively, will have a higher proportion of girl births.

He tested this with data from the **National Longitudinal Study of Adolescent Health (Add Health)**, in which interviewers rated each respondent's physical attractiveness on a 1–5 scale (1 = very unattractive, 5 = very attractive). To avoid confounds from family-size decisions, the analysis was restricted to *first-born* children only. Total sample: $n \approx 2{,}972$.

1.2 The raw data

Table 1: Proportion of sons by parental attractiveness rating (Add Health). Sample sizes inferred from Kanazawa's reported proportions and total $n = 2{,}972$.

Rating	Label	n (approx.)	% sons	% daughters
1	Very unattractive	68	52	48
2	Unattractive	246	53	47
3	About average	1532	52	48
4	Attractive	996	51	49
5	Very attractive	130	44	56
1–4	Combined	2842	52	48

Kanazawa compared category 5 with categories 1–4 and found

$$56\% - 48\% = 8 \text{ percentage points more daughters} \quad (t = 2.44,\ p \approx 0.015).$$

This was reported as "very attractive individuals are 26% less likely to have a son," and in media coverage as "36% more likely to have a daughter."

The "26% / 36%" confusion. These figures are *odds ratios*, not percentage-point differences. The odds ratio comparing sons in the 1–4 group to the 5 group is

$$\frac{52/48}{44/56} = \frac{0.5200}{0.3929} \approx 1.36,$$

meaning the *odds* of a son are about 36% higher in the 1–4 group. This is a valid statistic, but it does not mean the probability of a daughter increases by 36 percentage points. The actual probability difference is 8 p.p.

Exercise 1 (Odds ratios vs. probability differences). *(a) Verify the odds ratio calculation above.*

(b) If $P(\text{daughter} \mid \text{unattractive}) = 0.48$ and the odds ratio for daughters (attractive vs. unattractive) is 1.36, what is $P(\text{daughter} \mid \text{attractive})$?

(c) Show that the answer to (b) is approximately 0.56, consistent with the reported data.

2 Gelman's critique I: multiple comparisons

2.1 How many tests were really done?

Kanazawa chose the comparison "category 5 vs. categories 1–4." But with five ordered categories, there are four possible binary splits (Table 2). In addition, attractiveness was measured at three waves but only one was used. Across splits and waves, there are roughly $4 \times 5 \approx 20$ plausible comparisons. At $\alpha = 5\%$, we expect $0.05 \times 20 = 1$ false positive even under the global null.

Table 2: All possible binary splits of five ordered categories.

"High" group	"Low" group
{5}	{1,2,3,4}
{4,5}	{1,2,3}
{3,4,5}	{1,2}
{2,3,4,5}	{1}

2.2 The correct pre-specified test

Gelman (2007) recommended fitting a linear regression using attractiveness as a continuous predictor, standardised to mean 0 and SD = 0.5 (so the coefficient is comparable to a binary variable). The result is

$$\hat{\beta} = 0.047, \qquad \mathrm{SE}(\hat{\beta}) = 0.043, \qquad t = 1.09, \qquad p \approx 0.28,$$

with 95% confidence interval

$$\hat{\beta} \pm 1.96 \times \mathrm{SE} = [-3.9\%,\ +13.3\%].$$

This is entirely non-significant. The only "significant" result came from cherry-picking the most extreme category comparison.

Exercise 2 (Standardising the predictor)**.** *Gelman divides by* two *standard deviations (not one) to make the coefficient directly comparable to a binary variable.*

(a) Suppose the mean attractiveness is 3.2 and the SD is 0.8. What is the standardised attractiveness of a person rated 5?

(b) Interpret $\hat{\beta} = 0.047$ in plain English under this standardisation.

(c) Re-derive the 95% CI using the Normal approximation.

3 Gelman's critique II: Type M and Type S errors

3.1 What effects should we expect?

The sex-ratio literature gives us a strong empirical constraint: virtually every known environmental or parental factor shifts the sex ratio by at most 1–2 percentage points under normal conditions. The standard error of a proportion from a sample of $n = 2000$ is

$$\mathrm{SE} = \frac{0.5}{\sqrt{n}} = \frac{0.5}{\sqrt{2000}} \approx 1.1\%.$$

Statistical significance at $\alpha = 5\%$ (two-sided) requires $|\hat{\beta}| \geq 1.96 \times \text{SE}$. With SE $= 4.3\%$ the threshold is

$$|\hat{\beta}|_{\min} = 1.96 \times 4.3\% \approx 8.4\%.$$

But any plausible true effect is around 0.3–1 p.p. Any significant result must therefore be at least $8.4/1 \approx 8$ times larger than the truth.

3.2 Type M and Type S errors

Gelman & Tuerlinckx (2000) define two errors beyond the classical Type I / Type II:

- **Type M (magnitude) error**: the estimated $|\hat{\beta}|$ is much larger than the true $|\beta|$.
- **Type S (sign) error**: $\hat{\beta}$ has the opposite sign to β.

Let $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2)$ with $\sigma = 4.3\%$. We reject when $|\hat{\beta}| > 1.96\sigma = 8.4\%$. The power is

$$\pi(\beta) = \Phi\left(\frac{-8.4\% - \beta}{\sigma}\right) + 1 - \Phi\left(\frac{8.4\% - \beta}{\sigma}\right).$$

Table 3: Power and inflation for different true effect sizes. $\sigma = 4.3\%$, $\alpha = 5\%$ two-sided.

True β	Power $\pi(\beta)$	Inflation factor	P(wrong sign | sig.)
0	5%	—	50%
0.3 p.p.	$\approx 3\%$	$>27\times$	$\approx 40\%$
1 p.p.	$\approx 4\%$	$>8\times$	$\approx 40\%$
3 p.p.	$\approx 10\%$	$>2.8\times$	$\approx 24\%$

Exercise 3 (Power calculations). *(a) Using the formula above, verify the power for $\beta = 1\%$.*

(b) Suppose the true effect is $\beta = 0.3\%$. If the result is statistically significant, the estimate must exceed 8.4%. What is the inflation factor $\hat{\beta}_{\min}/\beta$?

(c) How large would n need to be to achieve 80% power for detecting $\beta = 1\%$ at $\alpha = 5\%$ (two-sided)? Recall: $n = (z_{\alpha/2} + z_{1-\pi})^2 \sigma_p^2 / \beta^2$ where $\sigma_p \approx 0.5$ for a proportion.

4 The People magazine cross-check

As an independent test, Gelman & Weakliem (2009) collected the sexes of children born to People magazine's "50 Most Beautiful People" lists for 1995–2000. The data were gathered in August 2007 from Wikipedia, the Internet Movie Database, and celebrities' personal webpages. Duplicates (celebrities appearing on multiple annual lists) were removed. A small number of entries were missing each year (2–6 per year).

One important caveat: unlike the Add Health analysis, *these data are not restricted to first-born children.* All children of each celebrity are included. This means the family-size confound that Kanazawa's design carefully avoided is potentially present here.

Population baseline: $\theta_0 = 48.5\%$ girls. Combined result: $157/329 = 47.7\%$, with SE $= 0.5/\sqrt{329} \approx 2.8\%$. The discrepancy of $47.7\% - 48.5\% = -0.8\%$ is in the *wrong* direction and nowhere near significant ($t \approx -0.28$). The encouraging results in 1995–96 (+7–8 p.p.) were purely due to chance; later years did not continue the pattern.

Table 4: Sex of children of People magazine's "50 Most Beautiful People," 1995–2000 (Gelman & Weakliem, 2009). Not restricted to first-borns.

Year	Girls	Boys	Total	% girls
1995	32	24	56	57.1%
1996	45	35	80	56.2%
1997	24	35	59	40.7%
1998	21	25	46	45.7%
1999	23	30	53	43.4%
2000	29	25	54	53.7%
Total	157	172	329	47.7%

Exercise 4 (Sequential inference). *(a) Using only the 1995 data (32 girls, 24 boys), compute a 95% CI for θ. Does it contain 0.485?*

(b) Add 1996: 77 girls, 59 boys combined. Re-compute. Does 0.485 still lie inside?

(c) Add all six years (157 girls, 172 boys). Does the CI contain 0.485?

(d) What does this illustrate about stopping data collection when results look favourable?

5 The conjugate Bernoulli model

5.1 Model specification

Model each birth as an independent Bernoulli trial:

$$X_i \mid \theta \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(\theta), \quad i = 1, \ldots, n,$$

where $\theta = P(\text{girl})$. The sufficient statistics are $g = \sum_i X_i$ (girls) and $n - g$ (boys). The likelihood is

$$p(\mathbf{x} \mid \theta) = \theta^g (1-\theta)^{n-g}.$$

5.2 Conjugate Beta prior and posterior

Choose $\theta \sim \text{Beta}(\alpha_0, \beta_0)$. By conjugacy the posterior is

$$\theta \mid \mathbf{x} \;\sim\; \text{Beta}\big(\alpha_0 + g,\;\; \beta_0 + (n-g)\big),$$

with posterior mean

$$\mathbb{E}[\theta \mid \mathbf{x}] = \frac{\alpha_0 + g}{\alpha_0 + \beta_0 + n}.$$

This is a weighted average of the prior mean $\mu_0 = \alpha_0/(\alpha_0+\beta_0)$ and the MLE $\hat{\theta} = g/n$:

$$\mathbb{E}[\theta \mid \mathbf{x}] = \frac{\alpha_0 + \beta_0}{\alpha_0 + \beta_0 + n}\,\mu_0 \;+\; \frac{n}{\alpha_0 + \beta_0 + n}\,\hat{\theta}.$$

5.3 Choosing the prior

We consider three priors, all centred at $\theta_0 = 0.485$ (the population girl-birth rate):

The informative prior Beta(485, 515) is equivalent to having already observed 1,000 births at the population rate. This is justified given the large literature showing sex-ratio effects of at most $\approx 1\%$.

Table 5: Prior specifications.

Prior type	α_0	β_0	Prior SD	Rationale
Flat (uniform)	1	1	$\approx 28.9\%$	No prior knowledge
Weakly informative	48.5	51.5	$\approx 3.5\%$	Baseline rate known
Informative (Gelman)	485	515	$\approx 1.1\%$	Full sex-ratio literature

5.4 Worked calculation: Add Health "very attractive" group

Data: $n = 130$ first-born children of "very attractive" parents, of whom $g = 73$ are girls ($\hat{\theta} = 56\%$).

Flat prior $\text{Beta}(1, 1)$:

$$\begin{aligned} \text{Posterior} &= \text{Beta}(1 + 73,\ 1 + 57) = \text{Beta}(74,\ 58) \\ \mathbb{E}[\theta \mid \mathbf{x}] &= 74/132 = 56.1\% \\ 95\%\ \text{credible interval} &\approx [47.2\%,\ 64.7\%] \end{aligned}$$

Informative prior $\text{Beta}(485, 515)$:

$$\begin{aligned} \text{Posterior} &= \text{Beta}(485 + 73,\ 515 + 57) = \text{Beta}(558,\ 572) \\ \mathbb{E}[\theta \mid \mathbf{x}] &= 558/1130 = 49.4\% \\ 95\%\ \text{credible interval} &\approx [46.4\%,\ 52.4\%] \end{aligned}$$

With the informative prior the data ($n = 130$) are too sparse to move the posterior appreciably away from the prior ($n_0 = 1000$). The posterior mean of 49.4% is barely above the baseline of 48.5%, and the credible interval comfortably contains 48.5%.

Exercise 5 (Conjugate update). *(a) Verify the posterior parameters for both cases above.*

(b) Show algebraically that the posterior mean is a weighted average of the prior mean and the MLE, and identify the weights.

(c) For the informative prior, compute $P(\theta > 0.485 \mid \mathbf{x})$ *using the Normal approximation: if* $\theta \sim \text{Beta}(a, b)$ *with* $a + b$ *large, then*

$$\theta \mathrel{\dot\sim} \mathcal{N}\!\left(\frac{a}{a+b},\ \frac{ab}{(a+b)^2(a+b+1)}\right).$$

(d) How many additional observations (still at the 56% girl rate) would be needed before the 95% credible interval under the informative prior excludes 48.5%?

Exercise 6 (People magazine data). *Use the combined People magazine data:* $g = 157$ *girls,* $n = 329$. *Note these are* all *children, not just first-borns.*

(a) Compute the posterior under the flat prior.

(b) Compute the posterior under the informative prior $\text{Beta}(485, 515)$.

(c) Compare the posterior means from (a) and (b) with the baseline $\theta_0 = 48.5\%$ *and interpret.*

(d) Does the 95% credible interval under either prior exclude $\theta_0 = 48.5\%$*? What would you conclude?*

6 The conjugate Normal model

6.1 Parameter of interest

Following Gelman, Hill & Vehtari (2020, §9.4), we work directly with the difference in proportions. Define

$$\theta = P(\text{girl} \mid \text{beautiful parents}) - P(\text{girl} \mid \text{other parents}).$$

We are interested in whether $\theta > 0$. All estimates and standard errors below are on the percentage-point scale (so $\theta = 8$ means an 8 p.p. difference), which avoids clumsy expressions like 0.0025 and reduces rounding errors in the worked calculations.

6.2 Why is the standard error so large?

The observed $\hat{\theta}_{\text{data}} = 8\%$ with $\text{SE}_{\text{data}} = 3\%$. It is worth understanding why the SE is as large as 3 p.p. for a survey of 3000 people.

- A simple random sample of size 3000 estimates a proportion with standard error $\sqrt{p(1-p)/3000}$. For $p \approx 0.5$ this gives $\sqrt{0.5 \times 0.5/3000} = 0.009$, i.e. about 1 p.p. Good, but we are not estimating a single proportion.
- We are comparing two groups, each of roughly size 1500. The SE of a difference of two proportions is

  $$\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}.$$

  With $p_1 \approx p_2 \approx 0.5$ and $n_1 = n_2 = 1500$:

  $$\sqrt{2 \times \frac{0.5 \times 0.5}{1500}} = \sqrt{\frac{0.5}{1500}} \approx 0.018 \quad (1.8 \text{ p.p.}).$$

  Under optimal conditions we can estimate the difference to within about 2 p.p.
- In reality the groups are far from equal: only about 10% of respondents were rated "very attractive" (category 5). The actual SE is therefore

  $$\sqrt{\frac{0.5 \times 0.5}{300} + \frac{0.5 \times 0.5}{2700}} = \sqrt{0.000833 + 0.0000926} \approx 0.03 \quad (3 \text{ p.p.}).$$

- Estimating a difference to ±2–3 p.p. is perfectly adequate for many purposes. For the sex ratio, however, we need precision on the scale of *0.1 p.p.* — more than an order of magnitude better. The survey is hopelessly underpowered for this question.

6.3 Prior estimate and standard error

We express prior knowledge as a Normal distribution on θ:

$$\theta \sim \mathcal{N}(0,\ \tau^2), \quad \tau = 0.25\%.$$

The prior mean of zero says that before seeing the data we have no reason to expect beautiful parents to have an elevated or depressed girl birth rate. The prior SD of 0.25% encodes the scientific knowledge that it is highly implausible for the true θ to exceed +0.5% or fall below −0.5%. Notice that the data SE (3%) is more than *ten times* the prior SD (0.25%): the prior is far more informative than the data.

6.4 Conjugate Normal–Normal update

The likelihood for θ given the data estimate is

$$\hat{\theta}_{\text{data}} \mid \theta \;\sim\; \mathcal{N}(\theta,\, \text{SE}_{\text{data}}^2), \quad \text{SE}_{\text{data}} = 3\%.$$

With the Normal prior above, conjugacy gives

$$\theta \mid \hat{\theta}_{\text{data}} \;\sim\; \mathcal{N}\big(\mu_{\text{post}},\, \sigma_{\text{post}}^2\big)\,,$$

where

$$\frac{1}{\sigma_{\text{post}}^2} = \frac{1}{\tau^2} + \frac{1}{\text{SE}_{\text{data}}^2}, \qquad \mu_{\text{post}} = \sigma_{\text{post}}^2 \cdot \frac{\hat{\theta}_{\text{data}}}{\text{SE}_{\text{data}}^2}.$$

The posterior mean is the shrinkage estimator

$$\mu_{\text{post}} = \hat{\theta}_{\text{data}} \cdot \frac{\tau^2}{\tau^2 + \text{SE}_{\text{data}}^2}.$$

6.5 Worked calculation (ROS §9.4)

Prior: $\tau = 0.25\%$. Data: $\hat{\theta}_{\text{data}} = 8\%$, $\text{SE}_{\text{data}} = 3\%$.

$$\frac{1}{\sigma_{\text{post}}^2} = \frac{1}{(0.0025)^2} + \frac{1}{(0.03)^2} = 160{,}000 + 1{,}111 = 161{,}111$$

$$\sigma_{\text{post}} = \frac{1}{\sqrt{161{,}111}} \approx 0.0025 \quad (0.25\%)$$

$$\mu_{\text{post}} = \frac{1}{161{,}111} \cdot \frac{0.08}{(0.03)^2} = \frac{1}{161{,}111} \cdot 88.9 \approx 0.00055 \quad (0.06\%)$$

So $\hat{\theta}_{\text{Bayes}} = 0.06\%$ with $\text{SE}_{\text{Bayes}} = 0.25\%$. The Bayes estimate is less than one-tenth of a percentage point — far smaller than its own standard error. The data tell us essentially nothing about variation in the sex ratio.

The same result via the shrinkage formula:

$$\mu_{\text{post}} = 8\% \times \frac{(0.25\%)^2}{(0.25\%)^2 + (3\%)^2} = 8\% \times \frac{0.0625}{0.0625 + 9} = 8\% \times 0.00692 \approx 0.055\%.$$

6.6 Comparison of prior scales

The prior SD of 0.25% is the scientifically appropriate choice, but it is instructive to see how sensitive the conclusion is.

Only if you are willing to entertain prior effects as large as ±3 p.p. (twelve times what the literature supports) does the data push the posterior to something notably positive. The conclusion is robust to any scientifically defensible prior.

Note on the Cauchy prior. In the published paper Gelman uses a Cauchy(0, 0.25%) prior rather than a Normal, because the Cauchy has heavier tails. For this course the Normal prior is sufficient; it gives the same qualitative conclusion and is algebraically tractable.

Exercise 7 (Normal–Normal update)**.** *(a) Derive the conjugate posterior formulas from scratch. Start from Bayes' theorem $p(\theta \mid \hat{\theta}) \propto p(\hat{\theta} \mid \theta)\, p(\theta)$ and complete the square in the exponent.*

Table 6: Posterior under various prior SDs. Data: $\hat{\theta} = 8\%$, $\text{SE}_{\text{data}} = 3\%$.

Prior τ	Shrinkage factor	μ_{post}	σ_{post}	$P(\theta > 0)$
$\tau = 0.25\%$ (ROS)	0.007	0.06%	0.25%	$\approx 59\%$
$\tau = 1\%$	0.100	0.8%	1.0%	$\approx 79\%$
$\tau = 3\%$ (= SE)	0.500	4.0%	2.1%	$\approx 97\%$
$\tau = 100\%$ (diffuse)	≈ 1	8.0%	3.0%	$\approx 996\%$

(b) Verify the worked calculation above using the shrinkage formula directly: compute μ_{post} for $\tau = 0.25\%$, $\text{SE}_{data} = 3\%$, $\hat{\theta} = 8\%$.

(c) The shrinkage factor is $\tau^2/(\tau^2 + \text{SE}^2)$. For what value of τ does the posterior mean equal exactly half the data estimate (shrinkage factor $= 0.5$)?

(d) Show that the precision of the posterior equals the sum of the prior and likelihood precisions: $1/\sigma^2_{post} = 1/\tau^2 + 1/\text{SE}^2$. Why is this the natural formula?

Exercise 8 (Sensitivity analysis)**.** *(a) Compute $P(\theta > 0)$ for $\tau \in \{0.1\%, 0.25\%, 1\%, 3\%, 10\%\}$. Tabulate and plot the results. At what τ does the probability first exceed 75%?*

(b) Verify the entry $P(\theta > 0) \approx 59\%$ for $\tau = 0.25\%$ using the standard Normal CDF.

(c) ROS notes that the data SE (3%) is more than ten times the prior SD (0.25%). Show this implies the shrinkage factor is approximately $\tau^2/\text{SE}^2 \approx 0.007$, i.e. the data contribute less than 1% of the weight in the posterior mean.

(d) Comment on whether the conclusion is sensitive to the choice of prior in this example.

7 Summary

The chain of problems in Kanazawa (2007):

1. *Arithmetic.* An odds ratio (1.36×) was reported as a probability change ("26% less likely"), then inflated by the media to "36% more likely to have a daughter."
2. *Multiple comparisons.* One of approximately 20 possible tests was reported — the one that happened to reach $p < 0.05$. The pre-specified regression on the continuous predictor is not significant ($p \approx 0.28$).
3. *Insufficient power.* The study can only detect effects of $\geq 8.4\%$; realistic sex-ratio effects are $\leq 1\%$. Any significant result must be a large overestimate (Type M error), and has a substantial chance of pointing in the wrong direction (Type S error).
4. *Bayesian conclusion.* A prior encoding the sex-ratio literature (SD = 0.25%) overwhelms the data (SE = 3%, more than ten times larger). The Bayes estimate is $\hat{\theta}_{\text{Bayes}} = 0.06\%$ with $\text{SE}_{\text{Bayes}} = 0.25\%$: the magnitude is smaller than its own standard error, and $P(\theta > 0) \approx 59\%$ — essentially no information.

The general lesson is about the interaction between power and the significance filter. When the standard error ($\sigma = 4.3\%$) is much larger than any plausible true effect ($\beta \leq 1\%$), the only estimates that pass the significance threshold are gross exaggerations. This is why prior

Table 7: Summary of analyses.

Analysis	Result	Conclusion
Kanazawa's comparison	$t = 2.44$, $p = 0.015$	"Significant"
Gelman's regression	$t = 1.09$, $p = 0.28$	Non-significant
95% CI for β	$[-3.9\%, +13.3\%]$	Consistent with zero
Bayesian (diffuse prior)	$P(\theta > 0) \approx 100\%$	Strongly positive
Bayesian ($\tau = 0.25\%$, ROS)	$\hat{\theta} = 0.06\%$, $P(\theta > 0) \approx 59\%$	No information
People magazine (all children)	47.7% girls ($<$ baseline)	No evidence

power calculations matter, why Bayesian priors informed by the literature are useful, and why a peer-reviewed publication does not guarantee that an estimated effect size is realistic.

Exercise 9 (Synthesis)**.** *(a) A new study collects data on 500 very attractive parents and finds 265 daughters (53%). Update the informative prior* $\text{Beta}(485, 515)$*. What is the posterior mean? Does the 95% credible interval exclude 48.5%?*

(b) How many observations would be needed for the posterior mean (under $\text{Beta}(485, 515)$*) to reach 50%, given a raw rate of 56% daughters?*

(c) Suppose a replication study has $n = 10{,}000$ *first-born children of "very attractive" parents, of whom 53% are daughters. Compute the frequentist 95% CI and the Bayesian 95% credible interval under both the flat and informative prior. Are the conclusions now different?*

(d) In your own words: why is it dangerous to report a statistically significant result from an underpowered study without discussing Type M and Type S errors?

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